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BCACACS 203

**Second Semester B.C.A. Degree Examination, June/July 2025
(SEP)**

**(2024-25 Batch Onwards)
COMPUTATIONAL MATHEMATICS**

Time : 3 Hours

Max. Marks : 80

Note : 1) Answer **any ten** questions from Part – **A** and **one full** question from **each** Unit of Part – **B**.

2) **Scientific Calculator is allowed.**

PART – A

(10×2=20)

1. a) Find the relative error of the number 8.6 if both of its digits are correct.
- b) Write the formula of Newton-Raphson Method.
- c) Define interpolation and extrapolation.
- d) If Δ is the forward difference operator then $\Delta^3 Y_3 = ?$
- e) Write the Simpson's 3/8 rule for numerical integration.
- f) What is a power function ?
- g) Define square matrix. Give example.
- h) Define lower triangular and upper triangular matrix. Give example.
- i) Define symmetric and skew symmetric matrix. Give example.
- j) Write the Taylor's series for $y(x)$.
- k) Write Euler's general formula.
- l) Write Milne's predictor formula.

PART – B

Unit – I

2. a) Use the method of iteration to find a positive root between 0 and 1 for the equation $xe^x = 1$ (Carry out only 4 iterations).
- b) The population of a town in the decennial census was as given below. Estimate the population for the year 1925 using Newton's backward difference interpolation formula.

Year (x)	1891	1901	1911	1921	1931
Population (y) (in thousands)	46	66	81	93	101

- c) If $y_1 = 4, y_3 = 12, y_4 = 19, y_x = 7$, find x using Lagrange's interpolation formula. **(5+5+5)**

P.T.O.



3. a) Find the real root of the following equation $f(x) = x^3 - x - 1 = 0$ using bisection method (Carry out only 3 iterations).
- b) In the table below the value of y are consecutive terms of a series of which the number 21.6 is the 6th term. Find the first term of the series using Newton's Forward Difference interpolation formula.

X	3	4	5	6	7	8	9
Y	2.7	6.4	12.5	21.6	34.3	51.2	72.9

- c) Certain corresponding values of x and $\log_{10} x$ are (300, 2.4771), (304, 2.4829), (305, 2.4843) and (307, 2.4871). Find $\log_{10} 301$ using Newton's Divided Difference formulae. (5+5+5)

Unit – II

4. a) Evaluate $I = \int_0^1 \frac{1}{1+x} dx$ by Simpson's 1/3 rule when $h=0.25$, correct to three decimal places.
- b) From the following table of values of x and y , obtain for $x = 1.2$ using Newton's forward difference formula.

X	1.0	1.2	1.4	1.6	1.8	2.0	2.2
Y	2.7183	3.3201	4.0552	4.9530	6.0496	7.3891	9.0250

- c) Determine the constants a and b by the method of least squares such as $y=ae^{bx}$, fit the following data.

x	2	4	6	8	10
y	4.077	11.084	30.128	81.897	222.62

(5+5+5)

5. a) The table below gives the temperature T (in $^{\circ}\text{C}$) and lengths l (in mm) of a heated rod. If $l = a_0 + a_1 T$, find the best values for a_0 and a_1 .

T	20	30	40	50	60	70
l	800.3	800.4	800.6	800.7	800.9	801.0

- b) Find the area bounded by the curve and x -axis from $x = 7.47$ to $x = 7.52$ from the following table using Trapezoidal rule.

X	7.47	7.48	7.49	7.50	7.51	7.52
f(x)	1.93	1.95	1.98	2.01	2.03	2.06



- c) From the following table of values of x and y, obtain for x = 6 using Newton's backward difference formula.

X	0	1	2	3	4	5	6
Y	6.9897	7.4036	7.7815	8.1291	8.4510	8.7506	9.0309

(5+5+5)

Unit – III

6. a) Solve the following equations using matrix inversion method.

$$3x + y + 2z = 3$$

$$2x - 3y - z = 3$$

$$x + 2y + z = 4$$

- b) Express the matrix $A = \begin{bmatrix} 1 & 7 & 8 \\ 6 & 2 & 9 \\ 5 & 4 & 3 \end{bmatrix}$ as a sum of symmetric and a skew-symmetric matrix.

- c) Show that the matrix $B = \begin{bmatrix} -2/3 & 1/3 & 2/3 \\ 2/3 & 2/3 & 1/3 \\ 1/3 & -2/3 & 2/3 \end{bmatrix}$ is orthogonal.

(5+5+5)

7. a) Use Gauss Seidal Method to solve the following system. Carry out 3 iterations.

$$10x_1 - 2x_2 - x_3 - x_4 = 3$$

$$-2x_1 + 10x_2 - x_3 - x_4 = 15$$

$$-x_1 - x_2 + 10x_3 - 2x_4 = 27$$

$$-x_1 - x_2 - 2x_3 + 10x_4 = -9$$

- b) Solve the following system of equations using Gauss Elimination method

$$5x - 2y + z = 4$$

$$7x + y - 5z = 8$$

$$3x + 7y + 4z = 10$$

- c) If $A = \begin{bmatrix} 2 & 3 & 2 \\ 1 & 0 & 0 \\ 2 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 & 1 \\ 2 & 3 & 4 \\ 1 & -2 & 3 \end{bmatrix}$ form AB and BA.

(5+5+5)



Unit – IV

8. a) Determine the value of y when $x = 0.1$ given that $y(0) = 1$ and $y' = x^2 + y$, $h = 0.05$, using Euler's modified method.
- b) Given $dy/dx = y - x$ where $y(0) = 2$ and $h = 0.1$. Find $y(0.1)$ using Runge-Kutta fourth order formula.
- c) Using Adams-Moulton formula, solve $y' = 1 + y^2$ where $y = 0$ when $x = 0$ and $h = 0.2$. Find $y(0.8)$. Given that $y(0.2) = 0.2027$, $y(0.4) = 0.4228$, $y(0.6) = 0.6841$.

(5+5+5)

9. a) From the Taylor's series for $y(x)$, find $y(0.1)$ correct to four decimal places if $y(x)$ satisfies $y' = x - y^2$ and $y(0) = 1$.
- b) Solve by Euler's method, the equations $dy/dx = x + y$, $y(0) = 0$, choose $h = 0.2$ and compute $y(0.4)$.
- c) Using Milne's formula solve $y' = 1 + y^2$ where $y(0) = 0$ and $h = 0.2$. Find $y(0.8)$ and $y(1.0)$. Given that $y(0.2) = 0.2027$, $y(0.4) = 0.4228$, $y(0.6) = 0.6841$.

(5+5+5)